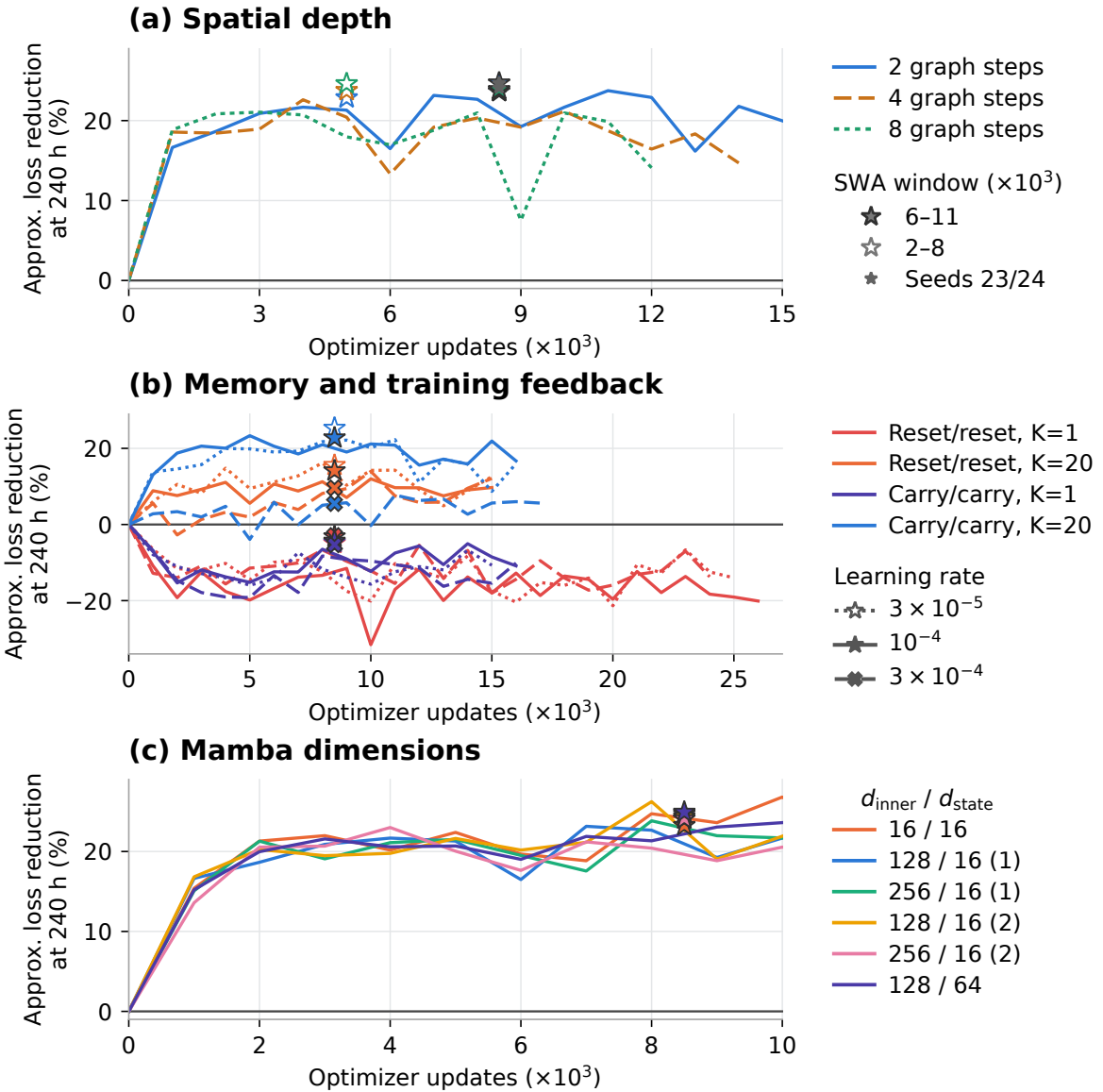
(a) Spatial depth
Approx. loss reduction at 240 h (%)
Optimizer updates (×10^3)
2 graph steps
4 graph steps
8 graph steps
SWA window (×10^3)
6–11
2–8
Seeds 23/24
(b) Memory and training feedback
Approx. loss reduction at 240 h (%)
Optimizer updates (×10^3)
Reset/reset, K=1
Reset/reset, K=20
Carry/carry, K=1
Carry/carry, K=20
Learning rate
3 × 10^−5
10^−4
3 × 10^−4
(c) Mamba dimensions
Approx. loss reduction at 240 h (%)
Optimizer updates (×10^3)
d_inner / d_state
16 / 16
128 / 16 (1)
256 / 16 (1)
128 / 16 (2)
256 / 16 (2)
128 / 64